$$= 20 + \frac{17-16}{2\times 17-16-15}\times C$$

$$= 20 + \frac{1}{34-31}\times 5$$

$$= 20 + \frac{5}{3} = 20 + 1.67 = 21.67$$

Mode = 21.67

(d) Graphical Location of Mode

The following are the steps to locate mode by graph

(i) Draw a histogram of the given distribution.

(ii) Join the rectangle corner of the highest rectangle (modal class rectangle) by a straight line to the top right corner of the preceding rectangle. Similarly the top left corner of the highest rectangle is joined to the top left corner of the rectangle on the right.

(iii) From the point of intersection of these two diagonal lines, draw a perpendicular line to the x –axis which meets at M.

(iv) The value of x coordinate of M is the mode.

Example 5.27

Locate the modal value graphically for the following frequency distribution

Class Interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency	5	8	12	7	5	3

Solution:

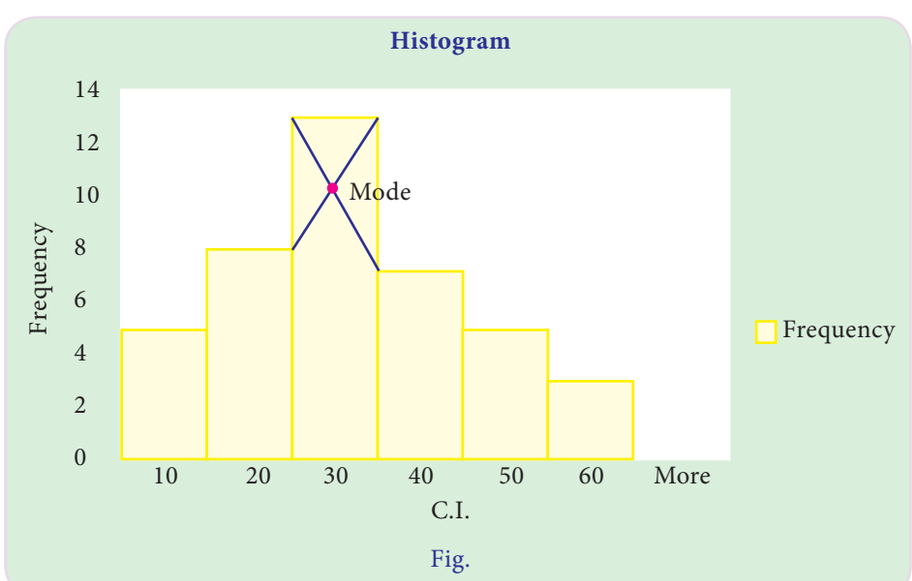


Fig.

Ch5_11th Stats.indd 140 2/3/2019 11:01:04 AM